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13	B	When the mass of the satellite is m and the mass of the planet is M, the radius of the orbit is r and the centripetal force on the satellite of mass m is F. $F = \frac{GMm}{r^2}$ The mass of a second planet is $3M$. A satellite of mass $2m$ orbits the planet with orbital radius $4r$. $F' = \frac{G(3M)(2m)}{(4r)^2}$ $= \frac{3}{8} \frac{GMm}{r^2} = 0.375 \frac{GMm}{r^2}$
14	D	Acceleration due to gravity experienced by the rock